

2023



Arizona State University Achieves Single-Digit Millisecond Latency for File Storage Using AWS Local Zones

Learn how Arizona State University centralized its IT operations on AWS by achieving single-digit millisecond latency using AWS Local Zones.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



93% decrease in latency
to 3-4 milliseconds



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Lowered total cost
of ownership



Overview

[Arizona State University](#) (ASU), a public research university located in the Phoenix metropolitan area, is a leader in educational innovation because it embraces technologically enhanced education and experiences for students and faculty. When moving its file-sharing applications from on-premises to the cloud, ASU needed a solution with low latency so that clients, typically Windows or Mac users handling MS documents, PDFs, and back-office file sharing, would not have to face wait times while working or collaborating on the cloud.

ASU has been using Amazon Web Services (AWS) since 2009 and has an “AWS First” strategy for public cloud adoption, meaning it prioritizes the latest in AWS technology. In 2022, ASU decided to use [AWS Local Zones](#)—a type of infrastructure deployment that places compute, storage, database, and other select AWS services close to large population and industry centers—to reduce latency and optimize connectivity for its applications in the cloud. AWS Local Zones in Phoenix, Arizona, brings AWS infrastructure to the same metro area as the university, helping ASU lower latency to resources. To move to AWS Local Zones, ASU worked with [Qumulo](#), an [AWS Partner](#) offering cloud-based file storage. Using AWS, ASU reduced latency for its file-sharing applications by 93 percent while supporting a collaborative approach in the community.



Opportunity | Using AWS Local Zones to Centralize Operations on the Cloud for ASU

ASU was established in 1885. In 2023, for the 9th straight year, it was ranked number one in innovation by [US News & World Report](#). ASU is at the forefront of using new AWS services and seeks to provide agility to its teams for rapid experimentation. Within the ASU charter is the assumption of fundamental responsibility for the economic, social, cultural, and overall health of the communities it serves, and it has taken steps to advance these goals using AWS.

For example, facilitated by AWS, ASU created the [Smart City Cloud Innovation Center](#) in 2018. “At the Smart City Cloud Innovation Center, ASU students accept challenges in the city, from homelessness to parking issues,” says John Rome, deputy chief information officer at ASU. In its first 5 years, the Smart City Cloud Innovation Center successfully completed 50 such challenges.



ASU has also used many AWS instructor-led trainings for its staff. A third of its Enterprise Technology team has achieved [AWS Certification](#), which validates technical skills and cloud expertise. The university uses over 150 AWS services, so AWS-certified staff means these services will be used more efficiently.

When ASU decided to move its file-sharing applications from on-premises to the cloud, AWS was its first choice. The most important aspect for ASU was maintaining the same latency for applications in the cloud as it had on premises. However, in the initial implementation, latency increased from no perceptible delay when accessing files to 40 milliseconds, which could cause users to associate the move to the cloud as a negative.

Solution | Achieving 3–4 Millisecond Latency Using AWS Local Zones

ASU migrated its file-sharing applications to [Amazon FSx](#)—used to launch, run, and scale feature-rich and highly performant file systems with just a few clicks. ASU moved over 400 TB of data to Amazon FSx in less than a year and initially ran the solution from the Oregon AWS Region. To reduce the latency of its solution, ASU chose to move its applications to the Phoenix AWS Local Zone, which was launched in 2022. However, at this time, Amazon FSx was not available in the Phoenix AWS Local Zone.

ASU turned to Qumulo for a file-storage solution. [Qumulo](#) provides an unstructured data solution in most AWS Regions and helps its customers manage petabytes of data with a modern, scale-out file system for high-performance workloads. ASU quickly performed a proof of concept with Qumulo, and its staff and users most sensitive to latency liked the results. The procurement process was fast because Qumulo is available on [AWS Marketplace](#), where businesses can find, test, buy, and deploy software that runs on AWS. ASU was able to cut the procurement time for this file storage solution, a lengthy and complicated process for a public institution, from 9 months to a matter of minutes.

ASU runs the Qumulo software on top of these resources to create file sharing for customers who require low latency. After successfully moving its solution to the Phoenix AWS Local Zone, ASU decreased latency to 3–4 milliseconds, a reduction of 93 percent.



To provide a reliable and secure connection from ASU data centers to the AWS Local Zone, ASU used [AWS Direct Connect](#), a cloud service that creates a dedicated network connection to AWS. “This solution was feasible for us because we used AWS Direct Connect,” says Nate Wilken, executive director of engineering for Enterprise Technology at ASU. “The encrypted data traveling over commodity internet would have resulted in high latency and security risk.”

The university is saving 6 percent on costs with its new solution and has achieved significant benefits in cloud cost offsets after decommissioning virtualization clusters. ASU has also achieved a lower cost of ownership. “We’re using the same tooling and automation pipelines,” says Wilken. “We can deploy locally without retaining a different solution for proximal technology deployment and can apply AWS strategy to improve customer experience.”

Outcome | Embracing Innovation for University and Community

In July 2023, AWS announced the launch of a second [Phoenix AWS Local Zone](#). In the wake of its success using AWS Local Zones, ASU plans to use the new Phoenix AWS Local Zone to facilitate the implementation of innovative technologies. For example, the university plans to improve developer productivity by deploying secure virtual desktop interfaces in the second Phoenix AWS Local Zone.

ASU is always looking for ways to innovate and embrace new technology. Recently, the university announced the launch of AI Acceleration, a new team of technologists dedicated to creating the next generation of artificial tools. The group is chartered with providing positive outcomes for students, paving new pathways for groundbreaking research, and developing the technical infrastructure for more efficient business operations. AI Acceleration is part of a larger initiative being led by ASU’s Enterprise Technology to drive strategy across the university that helps all students, faculty, and staff to make use of the advantages of AI to enhance their daily work.

“Technology is one of our superpowers that helps us meet the mission of the university, which includes demonstrating leadership in academic excellence and accessibility at scale,” says Rome. “Using AWS, we let someone else manage the context work of running servers so we can focus on core work involving learning and the student experience.”

About Arizona State University

Arizona State University is an American public research university. It is expanding academic opportunities for every type of learner and has been ranked number one for innovation 9 years in a row by US News & World Report.

AWS Services Used

Amazon FSx

Amazon FSx makes it easy and cost effective to launch, run, and scale feature-rich, high-p systems in the cloud.



[Learn more »](#)

AWS Local Zones

AWS Local Zones are a type of infrastructure deployment that places compute, storage, database, and other select AWS services close to large population and industry centers.

[Learn more »](#)

AWS Direct Connect

The AWS Direct Connect cloud service is the shortest path to your AWS resources. While in transit, your network traffic remains on the AWS global network and never touches the public internet.

[Learn more »](#)

AWS Certification

Validate technical skills and cloud expertise to grow your career and business.

[Learn more »](#)

More Education Customer Stories

EUROPE, MIDDLE EAST, & AFRICA



FHNW Creates Generative AI-Inspired Art Exhibit for...

The University of Applied Sciences and Arts Northwestern (FHNW) is one of Switzerland's largest universities. In

BELGIUM



collaboration with interactive museum for analog and digital technology Enter Technikwelt Solothurn, FHNW embarked on a generative artificial intelligence (AI) project using Amazon Web Services (AWS). With managed service Amazon SageMaker, says research associate Claudio Baonessa



2024

Amanote Optimizes Note-Taking App for Learners,...

Amanote can onboard new large university customers for its note-taking app in less than 2 minutes, scale to accommodate thousands of new users



2024

UNITED STATES



Enhancing School Safety and the Student Pickup...

Learn how educational technology company Pikmykid makes school dismissals simpler with its Dismissal Management software, developed on AWS.

UNITED KINGDOM



Ravensbourne
University London

Ravensbourne University Case Study

By using AWS Elemental for its video streaming services, Ravensbourne University has been able to reach a higher number of students with higher satisfaction. Ravensbourne is an innovative, industry-focused university located in the heart of London that specializes in the cross-section between creativity and technology. The



2019

1 2 3 4 5 6 ... 47

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.



Contact Sales

Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

Resources for AWS

- Getting Started
- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners

Developers on AWS

- Developer Center
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

Help

- Contact Us
- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |
- Español |
- Français |
- Italiano |
- Português |
- Tiếng Việt |
- Türkçe |
- Русский |
- ไทย |
- 日本語 |
- 한국어 |

